FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2023

TEST DATE: 15-01-2023

ANSWERS, HINTS & SOLUTIONS

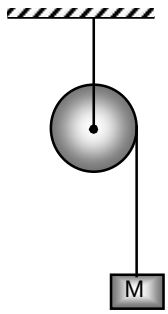
Physics

PART – A

SECTION – A

1. B

Sol. Applying conservation of angular momentum of system about centre of mass of disc



$$mvR = mv'R + \frac{2mR^2}{2}\omega$$

$$v' = R\omega$$

$$mvR = mv'R + mv'R$$

$$v' = \frac{v}{2} = \frac{\sqrt{2g\ell}}{2} = \sqrt{\frac{g\ell}{2}}$$

2. D

Sol. $I = neA(\mu_e + \mu_h)E$

$$= 1.0\times10^{6}\times1.6\times10^{-19}\times4\times10^{-4}(1.0)\times\frac{2}{2\times10^{-3}} = 6.4\times10^{-14}\ A$$

3. C

Sol. Basic concepts

4. C

Sol. $C = C_V + \frac{dW}{ndT}$

$$\int dW = \int_{T_0}^{3T_0}\alpha T dT - \int_{T_0}^{3T_0}\frac{5R}{2}dT \text{ for one mole}$$

$$W = (4\alpha T_0 - 5R)T_0$$

5. C

Sol. $\frac{dN}{dt} = R - \lambda N$

So $\int_0^N \frac{dN}{R-\lambda N} = \int_0^t dt \Rightarrow \ln(R-\lambda N)\Big|_0^N = -\lambda t$

So $\ln\frac{R-\lambda N}{R} = -\lambda t$

So $1-\frac{\lambda N}{R} = e^{-\lambda t} \Rightarrow N = \frac{R}{\lambda}\left(1-e^{-\lambda t}\right)$

6. C

Sol. Zener current will be maximum, when V = 15V

$15 - (i \times 2.5k) = 5$

$i = 4$ mA

$i_z = 3$ mA

7. B

Sol. $\frac{R_{Ge}}{R_{Be}} = \left(\frac{X}{9}\right)^{1/3}$

X = 72

Number of neutron = 72 – 32 = 40

8. B

Sol. frequency received by submarine

$n_1 = \left(\frac{1400+100}{1400}\right)\times 40\times 10^3 = \left(\frac{15}{14}\right)\times 40\times 10^3$

Frequency reflected by submarine

$n_2 = \left(\frac{1400}{1400-100}\right)n_1 = \left(\frac{15}{13}\right)\times 40\times 10^3$ Hz = $\frac{600}{13}$kHz

9. C

Sol. $f = \frac{2\pi(1-\cos\theta_C)}{4\pi}$

$f = \frac{2\pi}{4\pi}\left[1-\sqrt{1-\frac{1}{\mu^2}}\right] = \frac{1}{2}\left[1-\sqrt{1-\frac{1}{\mu^2}}\right]$

10. D

Sol. For ($r \le R$)

$E\times 4\pi r^2 = \int_0^r cx4\pi x^2 dx / \varepsilon_0$

So $E \propto r^2$

and For ($r \ge R$)

$E \propto \frac{1}{r^2}$

11. D

Sol. ${}_1^2H + {}_1^2H = {}_2^4He$

$BE_H = 1.1$ MeV

$BE_{He} = 7$ MeV

Q_{value} = Binding energy of product − Binding energy of reactants

$= (4 \times 7 - 4 \times 1.1)$ MeV = 23.6 MeV

$= 23.6 \times 1.6 \times 10^{-19} \times 10^6 = 37.76 \times 10^{-23}$

12. C

Sol. $B = \dfrac{\mu_0 dI}{2\pi R} = \dfrac{\mu_0 I d_0}{(2\pi R)(2\pi R)} = \dfrac{\mu_0 I d_0}{4\pi^2 R^2}$

13. D

Sol. After long time,

$I_A = \dfrac{2V}{7R} = 20$ mA …(i)

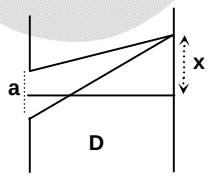
Immediately after the key is closed,

$I = \dfrac{V}{3R} = \dfrac{70}{3}$ mA

14. C

Sol. For 1st secondary minima $\dfrac{ax}{D} = \lambda_1$

For 1st secondary maxima $\dfrac{ax}{D} = \dfrac{3\lambda_2}{2}$



$\lambda_1 = \dfrac{3}{2}\lambda_2$

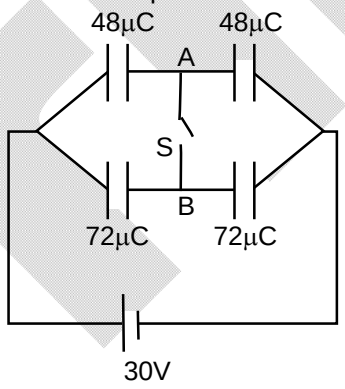
$\lambda = 440$ Å

15. D

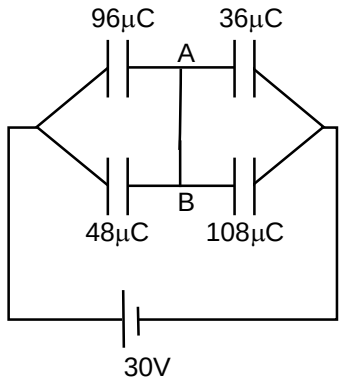
Sol. Basic Concepts

16. A

Sol. When S is open



When S is closed



So charge flown = 60 μC

From A to B

17. B

Sol. Basic concept

18. C

Sol. $E2\pi r = \dfrac{d}{dt}(B\pi r^2)$

$E = \dfrac{r}{2}\dfrac{dB}{dt}$

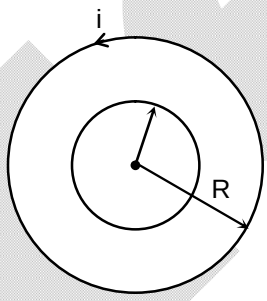
$\tau = QEr = \dfrac{Qr^2}{2}\dfrac{dB}{dt}$

19. B

Sol. $B = \dfrac{\mu_0 i}{2R}$

$\phi = B\pi r^2 = \dfrac{\mu_0 i \pi r^2}{2R}$

Mutual inductance, $M = \dfrac{\phi}{i} = \dfrac{\mu_0 \pi r^2}{2R}$



20. D

Sol. Rate of cooling

$-\dfrac{dT}{dt} = \dfrac{\varepsilon\sigma A(T^4 - T_s^4)}{mC} = r$

$\dfrac{r_{sphere}}{r_{cube}} = \dfrac{m_{cube}}{m_{sphere}} = \dfrac{\rho a^3}{\rho\dfrac{4}{3}\pi r^3}$

given, $4\pi r^2 = 6a^2$

$\dfrac{r_{sphere}}{r_{cube}} = \dfrac{3\left(\dfrac{4\pi}{6}\right)^{3/2} r^3}{4\pi r^3} = \sqrt{\dfrac{\pi}{6}}$

SECTION – B

21. 6

Sol. Separation between two maxima $\beta = \dfrac{\lambda D}{d}$

22. 40

Sol. $v_0^2 = 4v_0^2 - (2)(\mu g)(d) \Rightarrow 2\mu gd = 3v_0^2$

$v_0^2 = v^2 - (2)(5\mu g)(d) \Rightarrow v^2 = v_0^2 + 15V_0^2$

So $v = 4v_0$

23. 1

Sol. $\dfrac{2v}{2(\ell_0 + 1.2d)} = \dfrac{3v}{4(\ell_c + e)}$

So $4(23+e)=3(30+2e)$

So e = 1 cm

24. 9

Sol. $E_0 = CB_0$

$= 3\times10^8\times3\times10^{-8}$ = 9 V/m

25. 30

Sol. $\frac{1}{f_e}=-2\frac{1}{f_\ell}+\frac{1}{f_m}$,

Where $\frac{1}{f_\ell}=\left(\frac{4}{3}-1\right)\left(\frac{1}{40}\right)=\frac{1}{120}$

So $\frac{1}{f_e}=-2\frac{1}{120}-\frac{1}{20}=-\frac{1}{15}$cm

So R = 30 cm

26. 2000

Sol. $(100-v)10+(200-v)20+(50-v)40=0$

So v = 100 volt

$Q = 40\times50 = 2000\ \mu C$

27. 24

Sol. $\frac{dV}{dt}=\frac{\pi\Delta pr^4}{8\eta\ell}$

$Q=\frac{\pi P_0 r^4}{8\eta\ell}$

Now, $Q'=\frac{\pi3P_0(2r)^4}{8\eta\times2\ell}=\frac{3\pi P_0r^4}{\eta\ell}=24Q$

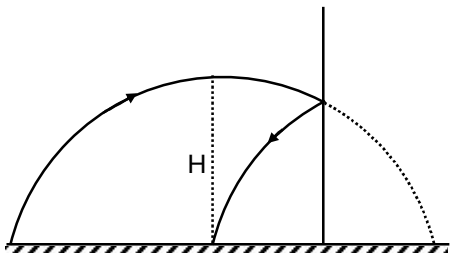
28. 4

Sol. $y=x\tan\theta\left(1-\frac{x}{R}\right)\Rightarrow H=\frac{R}{2}\tan\theta\left(1-\frac{1}{2}\right)$

$\Rightarrow\ H=\frac{R\tan\theta}{4}$

$y=\frac{3R}{4}\tan\theta\left(1-\frac{3R}{4R}\right)$

$=\frac{3R}{16}\tan\theta=\frac{3R}{16}\frac{4H}{R}=\frac{3H}{4}$



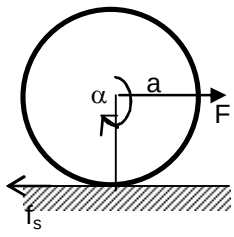
29. 16

Sol. $a = \alpha R$

$F - f_S = ma$ …(i)

$f_SR=\frac{mR^2\alpha}{2}$

$f_S=\frac{ma}{2}$ ..(ii)



$$F = \frac{3ma}{2}$$

$$\Rightarrow a = \frac{2F}{3m} \Rightarrow f_S = \frac{ma}{2} = \frac{F}{3}$$

30. 2

Sol. In steady state, v = constant

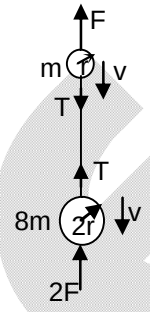
$9\ mg = 3F$

$\Rightarrow F = 3\ mg$ … (i)

$T + mg = F$

$T = F - mg = 3mg - mg = 2mg$

$T = 2mg$



Chemistry

PART – B

SECTION – A

31. B

Sol. $P_1V_1 = P_2V_2$

$10m \times (4) \times A = (10 + h)m \times h \times A$

$h = 3.06$ m

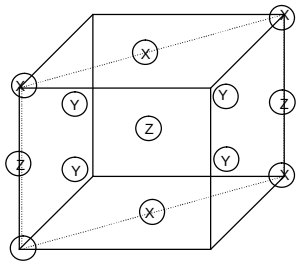
32. C

Sol. Dithionous acid $\rightarrow H_2S_2O_4$ $S = +3$

Dithionic acid $\rightarrow H_2S_2O_6$ $S = +5$

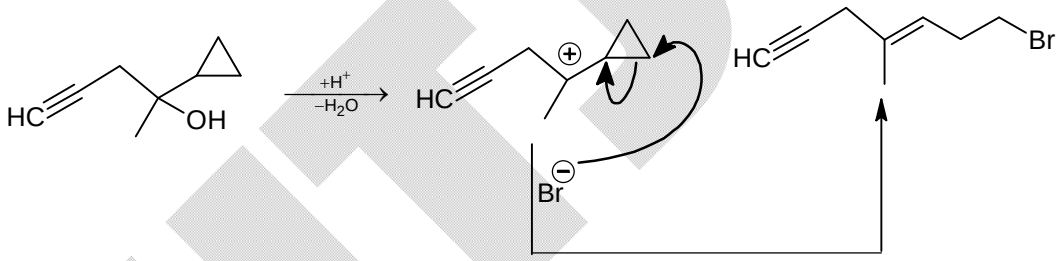
33. C

Sol.



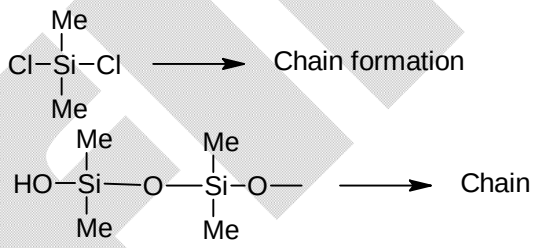
34. D

Sol.



35. B

Sol.



36. C

Sol. $LiClO_4$ is more soluble than LiF.

37. D

Sol. Factual

38. B

Sol. BF_3 is electron deficient and is used to generate electrophile in F.C. reactions. Therefore, acts as catalyst in industrial processes.

39. B

Sol. Uridine is glycosylated pyrimidine-containing uracil attached to ribose ring.

$\text{Ribose}(\text{sugar}) + \text{Uracil}(\text{base}) \longrightarrow \text{Nucleoside}$

40. A

Sol. Degenerate orbitals $= n^2$

$= (5)^2 = 25$

41. D

Sol. $\Delta S \rightarrow$ State function $\Delta S = \Delta S_1 + \Delta S_2$

$\Delta S = \frac{q}{T}$ since q = same; smaller temperature, higher entropy.

42. D

Sol. $Ti^{+3} \longrightarrow 3d^1 \longrightarrow t_{2g}^1$

$Cr^{+3} \longrightarrow 3d^3 \longrightarrow t_{2g}^3$

$Mn^{+3} \longrightarrow 3d^4 \longrightarrow t_{2g}^3 eg^1$

$Co^{+3} \longrightarrow 3d^6 \longrightarrow t_{2g}^4 eg^2$

43. A

Sol. Factual

44. B

Sol. MnO_2 oxidises unhindered primary allylic and benzylic alcohols to aldehyde.

45. C

Sol. $\frac{1}{x_o - a} - \frac{1}{x_o} = k_2 t$

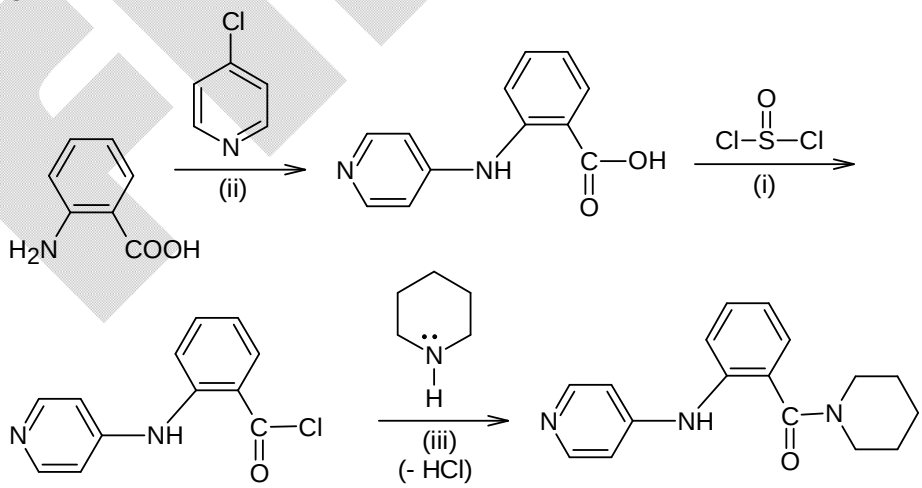
Rearranging

$\frac{a}{x_o - a} = x_o k_2 t$

$\therefore \frac{a}{x_o - a}$ vs. t gives $x_o k_2$ as slope.

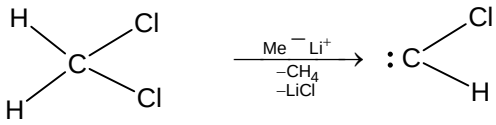
46. C

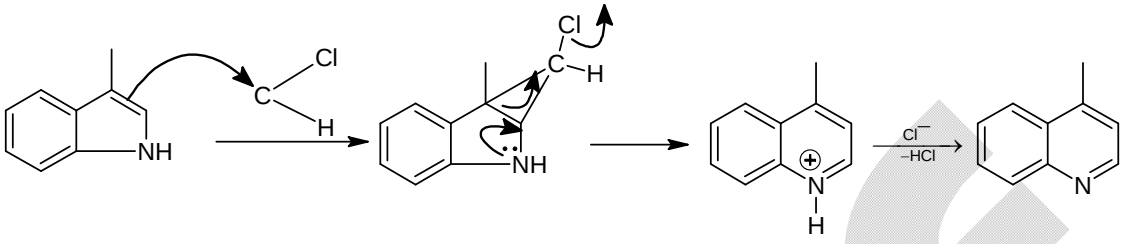
Sol.



47. A

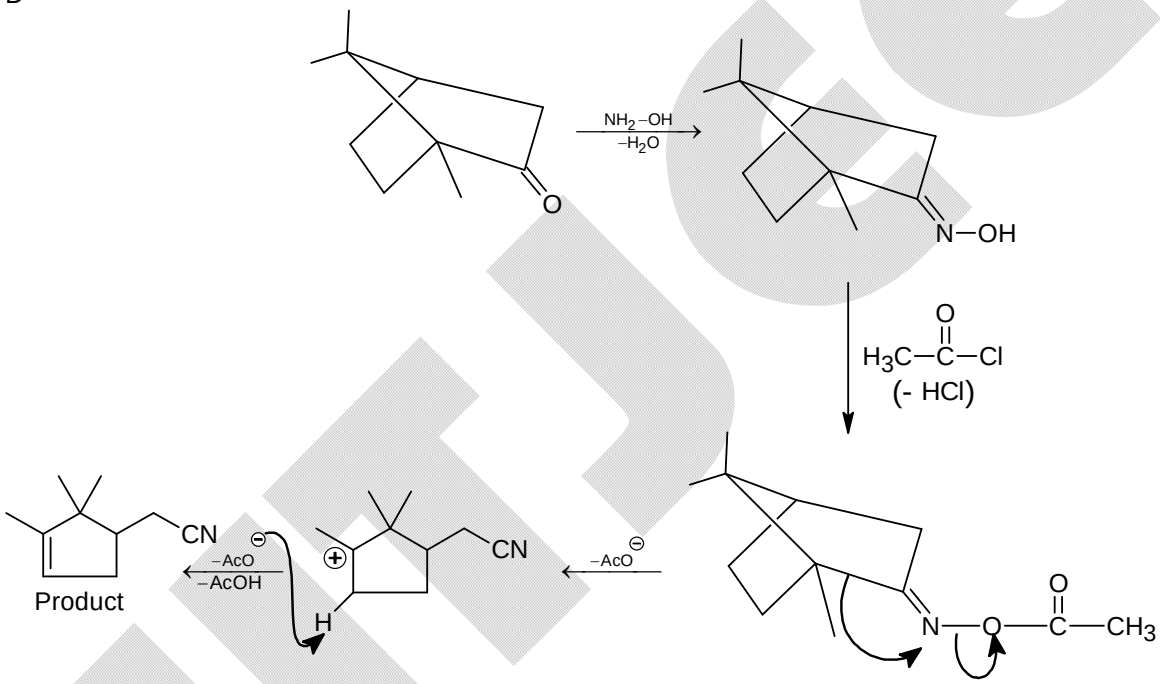
Sol.



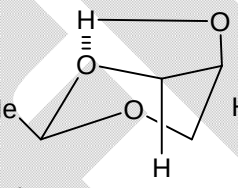


48. D

Sol.



49. B

Sol. 

Hydrogen bonding and as 1-3 dihedral interaction between methyl and hydrogen.

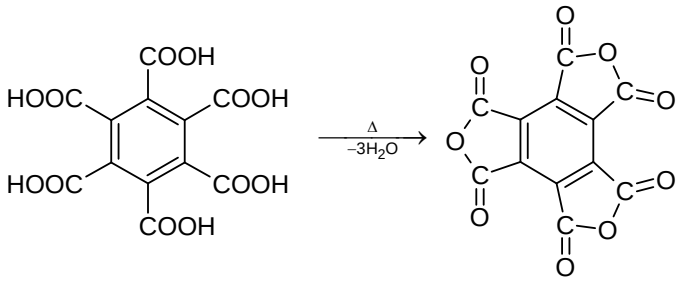
50. B

Sol. $x_A = \frac{0.744 + 0.359Z_{eff}}{r_A^2} \leftarrow$ Alfred – Rochow scale

SECTION – B

51. 13

Sol.



DOU = 13

52. 2

Sol. $\left[H^+\right] = 10^{-4}$

$\therefore \left[Ca^{+2}\right] = \frac{10^{-4}}{2} M$

Weight of $Ca^{+2} = 20 \times 10^{-4}$ g in 1 L

$\therefore$ Weight of Ca^{+2} ion in 10^6 mL $H_2O = \frac{20 \times 10^{-4} \times 10^6}{10^3} = 2$

$\therefore$ Hardness = 2 ppm.

53. 1

Sol. Theoretical efficiency $= \frac{-\Delta G^o}{-\Delta H^o}$

$0.84 = \frac{-\Delta G^o}{285}$

$\Delta G^o = -0.84 \times 285$ kJ $= -nFE^o_{cell}$

$E^o_{cell} = 1.24$ V

54. 2

Sol. $Cd^{+2} + H_2S \rightleftharpoons Cd(s) + 2H^+$

Initial	0.1×10^{-3} moles		$0.08 \times 10 \times 10^{-3}$ moles
final	–		$0.8 \times 10^{-3} + 0.2 \times 10^{-3}$
			$= 10^{-3}$ moles

$\therefore \left[H^+\right]_{final} = \frac{10^{-3}}{100} \times 1000 = 0.01\ M = 10^{-2}\ M$

pH = 2.0

55. 8

Sol. For KCN: $\Delta T_f = i.K_f.m \Rightarrow 0.8 = K_f.0.2 \times 2$

$K_f = 2$ … (i)

$Hg(CN)_2 + mCN^- \longrightarrow Hg(CN)^{-m}_{m+2}$

Initial	0.1	0.2	–
Final	$(0.2 - 0.1\,m)$		0.1

Final effective molality = (0.2 – 0.1 m) + 0.1 + 0.2

= 0.5 – 0.1 m

New: $\Delta T_f = K_f.m$

$0.6 = 2 \times (0.5 - 0.1\,m)$

m = 2

4m = 8

56. 3

Sol. $XeO_2F_2 \longrightarrow sp^3d \quad \rightarrow x = 3$

$B_2H_6 \longrightarrow sp^3 \quad \rightarrow x = 6$

$[NiCl_4]^{-2} \longrightarrow sp^3 \quad \rightarrow x = 3$

$[PtCl_4]^{-2} \longrightarrow dsp^2 \quad \rightarrow x = 2$

$N(SiH_3)_3 \longrightarrow sp^2 \quad \rightarrow x = 2$

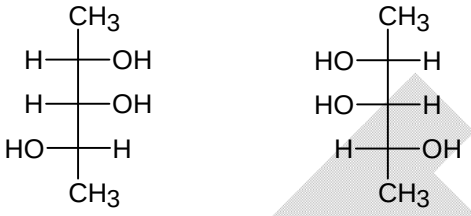
$H_3BO_3 \longrightarrow sp^2 \quad \rightarrow x = 2$

$\sum x = 3 + 6 + 3 + 2 + 2 + 2 = 18$

$\therefore \dfrac{\sum x}{6} = 3$

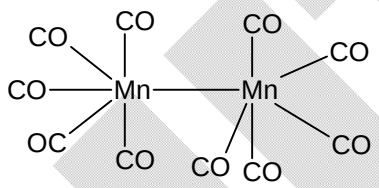
57. 2

Sol.



58. 0

Sol.



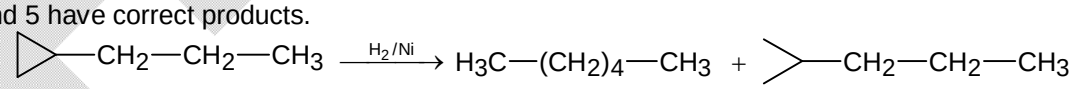
59. 6

Sol. Black coloured : PbS, CuS, HgS, Ag_2S, NiS and CoS.

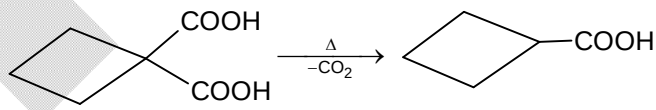
60. 6

Sol. 1 and 5 have correct products.

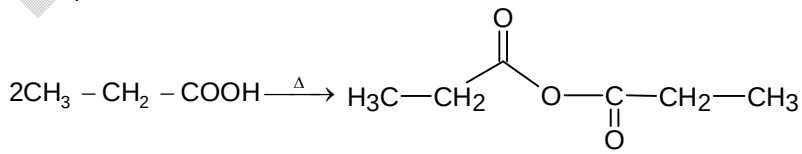
2.



3.



4.



Mathematics

PART – C

SECTION – A

61. B

Sol. $A = \{1, 2, 3, \ldots\ldots, 141\}$

$A \cap (B - C) = \{5, 11, 14, 17, 23, 26, 29, \ldots\ldots 134, 137\}$

Sum of elements = $(5 + 11 + 17 + \ldots\ldots 137) + (14 + 26 + 38 + \ldots.. 134)$

62. C

Sol. $x^2 - ax + 4 = 0$ must have atleast one positive root

$\because \quad \alpha\beta > 0$

$\therefore$ both roots must be positive

$\Rightarrow \quad a \geq 4$

63. B

Sol. $30(1 + x + x^2)^{30}(1 + 2x) = \sum_{r=0}^{60} r a_r x^{r-1}(1 + x + x^2)$

Equating coefficient of x^{20} on both sides we have,

$30(a_{20} + 2a_{19}) = 21a_{21} + 20a_{20} + 19a_{19}$

$\Rightarrow 21a_{21} = 10a_{20} + 41a_{19}$

64. C

Sol. Use $\int \left(xg'(x) + g(x)\right)dx = xg(x) + C_1$

65. C

Sol. $k = 2500 = 2^2.5^4$

$\therefore$ number of factor = 5

66. B

Sol. $b = -4, c = 9$

$2\sin^2\theta = 3\cos\theta$

$\Rightarrow 2\cos^2\theta + 3\cos\theta - 2 = 0$

$\cos\theta = \frac{1}{2} \quad \therefore \sin^2\theta = \frac{3}{4}$

$\therefore 4x^2 - 4y^2 + xy + 12x + \frac{3}{2}y + 9 = 0$

67. A

Sol. $r = \frac{r_1 r_2}{r_1 + r_2 + 2\sqrt{r_1 r_2}}$

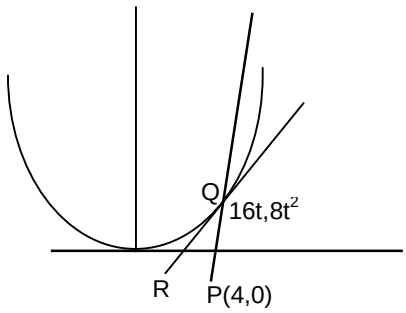
68. B

Sol. $A = \frac{1}{2}(4 - 8t)8t^2 = 16(t^2 - 2t^3)$

$\frac{dA}{dt} = 16(2t - 6t^2) = 0 \Rightarrow t = 0, \frac{1}{3}$

$\frac{d^2A}{dt^2} = 16(2 - 12t) < 0$ at $t = \frac{1}{3} \Rightarrow$ A is maximum

$\therefore Q\left(\frac{16}{3}, \frac{8}{9}\right)$ lies on the line $\therefore m = \frac{2}{3}$



69. A

Sol. M is (1, 10)

$\therefore$ N is (4, 16)

L is (–2, 4)

Equation of tangent to $(x + a)y = \lambda$ is

$$y = 2x + \frac{\lambda}{t} + 2(a - t);\ \lambda = -2t^2$$

$\therefore$ $2a - 4t = +8 \Rightarrow a - 2t = 4$

and $(a - 2)4 = \lambda = -2t^2$

$\therefore$ $(t + 2)^2 = 0$

$\Rightarrow$ $t = -2$

$\therefore$ $\lambda = -8$ & $a = 0$

70. B

Sol. $4a^2 - b^2 = 25$

Equation of auxiliary circle is $x^2 + y^2 = 8$

$\Rightarrow$ $a^2 = 8,\ b^2 = 7$

Also $a^2 + b^2 = A^2 - \lambda^2$

$\Rightarrow$ $\lambda^2 = 25 - 15 = 10$

$\therefore$ eccentricity of ellipse $e = \sqrt{1 - \frac{10}{25}} = \sqrt{\frac{3}{5}}$.

71. D

Sol. IF = $e^{x/2}\sqrt{\sin x + \cos x}$

$\therefore$ solution is $ye^{x/2}\sqrt{\sin x + \cos x} = \int e^{x/2}\left(\tan x + 2\sec^2 x\right)dx$

$ye^{x/2}\sqrt{\sin x + \cos x} = 2e^{x/2}\tan x + c$

$\because y\left(\frac{\pi}{4}\right) = 2^{3/4}$ $\therefore$ $c = 0$

$$y = \frac{2\tan x}{\sqrt{\sin x + \cos x}}$$

72. D

Sol. $\text{Arg}\left(\frac{z^4 - z}{z^2 - z}\right) = 0$

$\Rightarrow \text{Arg}\left(\frac{z(z-1)(z^2 + z + 1)}{z(z-1)}\right) = 0$; $z \neq 0, 1$

$\Rightarrow z^2 + z + 1$ is positive real number $(x + iy)^2 + x + iy + 1 > 0$

$\Rightarrow x^2 - y^2 + x + 1 > 0$ and $y(1 + 2x) = 0$

(i) $y = 0,\ x^2 + x + 1 > 0 \Rightarrow x \in R$

(ii) $x = -\frac{1}{2},\ y^2 < \frac{3}{4} \Rightarrow y \in \left(-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$.

73. C

Sol. $\frac{a}{a - \sqrt{3}} + \frac{b}{b - \sqrt{5}} + \frac{c}{c - \sqrt{7}} = 2$

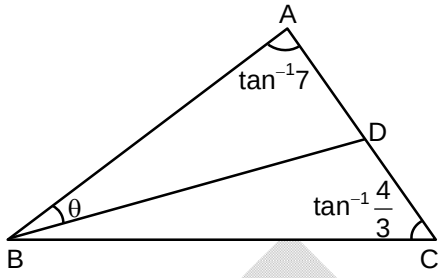
If $a = 2\sqrt{3}$, then $\frac{b}{b - \sqrt{5}} + \frac{c}{c - \sqrt{7}} = 0$

Possible value of $b = \frac{2\sqrt{5}}{3}$, $c = 2\sqrt{7}$.

74. D

Sol. $$\frac{\frac{1}{2}\sin\left(\tan^{-1}\frac{4}{3}\right)BD\sin\theta}{\frac{1}{2}\sin\left(\tan^{-1}7\right)BD\sin\left(\frac{\pi}{4}-\theta\right)}=2$$

$$\Rightarrow \frac{\frac{4}{5}\sin\theta}{\frac{7}{5\sqrt{2}}\left(\frac{\cos\theta-\sin\theta}{\sqrt{2}}\right)}=2 \Rightarrow \cot\theta=\frac{11}{7}$$



75. C

Sol. $P(A)=\frac{2}{5}$, $P(B)=\frac{3}{5}$

$P(D)=\frac{3}{20}$

$$6\times\frac{2}{5}x+\frac{3}{5}\times x=\frac{3}{20}\Rightarrow x=\frac{1}{20}$$

$$P\left(\frac{B}{D^C}\right)=\frac{P\left(B\cap D^C\right)}{P(D^C)}=\frac{\frac{3}{5}\times\frac{19}{20}}{\frac{85}{100}}=\frac{57}{85}$$

76. A

Sol. $$\frac{\sum_{i=1}^{n}x_if_i}{\sum_{i=1}^{n}f_i}=\frac{15a+25b+980}{26+a+b}=\frac{277}{9} \quad \ldots(1)$$

Also $$30+\frac{\frac{26+a+b}{2}-(a+b+3)}{11}\times 10=\frac{335}{11}$$

$\Rightarrow$ a + b = 19 ...(2)

From (1) and (2) a = 7, b = 12.

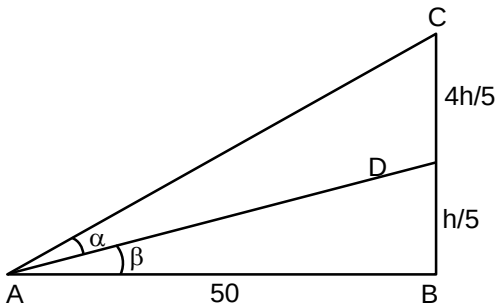
77. B

78. A

Sol. $\tan(\alpha+\beta)=\frac{h}{50}$

$$\Rightarrow \frac{h}{50}=\frac{\frac{8}{21}+\frac{h}{250}}{1-\frac{8}{21}\times\frac{h}{250}}=\frac{2000+21h}{5250-8h}$$

h = 25



79. B

Sol. $\boxed{3}\boxed{5}$ …… $= 9 \times 5 = 45$

__ $\boxed{3}\boxed{5}$ ____ $= 8 \times 5 = 40$

…… $= 8 \times 9 \times 9 \times 5 = 3240$

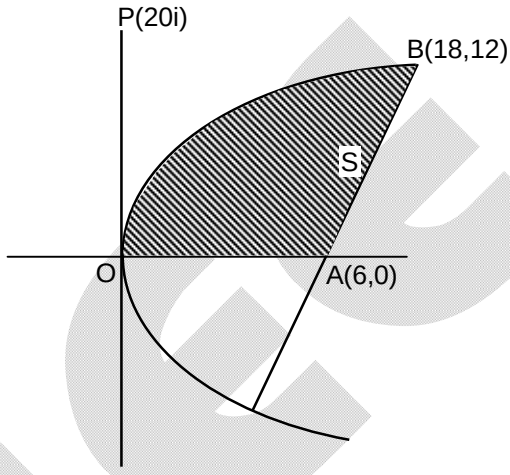
N = 3325

80. A

Sol. PB = $\sqrt{324+64} = \sqrt{388}$

PA = $\sqrt{36+400} = \sqrt{436}$

OP = 20



SECTION – B

81. 5

Sol. $$\sum_{m=1}^{15} \frac{\sin\left(\theta + m\frac{\pi}{18} - \left(\theta + (m-1)\frac{\pi}{18}\right)\right)}{\cos\left(\theta + m\frac{\pi}{18}\right)\cos\left(\theta + (m-1)\frac{\pi}{18}\right)}$$

$$= \sum_{m=1}^{15} \tan\left(\theta + m\frac{\pi}{18}\right) - \tan\left(\theta + (m-1)\frac{\pi}{18}\right)$$

$\Rightarrow \quad \tan\left(\theta + \frac{5\pi}{6}\right) - \tan\theta = 4 + 2\sqrt{3}$

$\Rightarrow \quad \cos\left(2\theta + \frac{5\pi}{6}\right) = 1$

$\Rightarrow \quad \theta = n\pi - \frac{5\pi}{12}$; $n \in I$.

82. 59

Sol. Squaring and adding

$1 + \sin^6 A + \cos^6 A + 2(\cos^4 A - \sin^4 A) = 1/4$

$\Rightarrow \quad 2 + 1 - 3\sin^2 A \cos^2 A + 2\cos 2A = 1/4$

$2 - 3/4 \sin^2 2A + 2\cos 2A = 1/4$

$\Rightarrow \quad 3\cos^2 2A + 8\cos 2A + 4 = 0$

$\cos 2A = -2/3$

$$\sin A = \sqrt{\frac{5}{6}}$$

$\therefore \quad \cos B = 2\left(\sqrt{\frac{5}{6}} - \frac{5}{6}\sqrt{\frac{5}{6}}\right) = \sqrt{\frac{5}{54}}$.

83. 13

Sol. $a = \frac{5\pi}{12}, b = \frac{\pi}{3}$

$$\int_{\pi/3}^{5\pi/12} \left[\frac{\tan x}{\sqrt{3}}\right] dx$$

$$= \int_{\pi/3}^{\tan^{-1}2\sqrt{3}} 1 + \int_{\tan^{-1}2\sqrt{3}}^{5\pi/12} 2dx = \cot^{-1}2\sqrt{3} = \operatorname{cosec}^{-1}13 .$$

84. 546

Sol. Let $A = I + B$ where $B = \begin{bmatrix} 0 & 0 & 0 \\ \sqrt{a} & 0 & 0 \\ a\sqrt{a} & \sqrt{b} & 0 \end{bmatrix}$, $B^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \sqrt{ab} & 0 & 0 \end{bmatrix}$, $B^3 = 0$ (Null)

$\therefore A^n = I + {}^nC_1B + {}^nC_2B^2$

$$\Rightarrow A^n = \begin{bmatrix} 1 & 0 & 0 \\ n\sqrt{a} & 1 & 0 \\ na\sqrt{a} + \frac{n(n-1)}{2}\sqrt{ab} & n\sqrt{b} & 1 \end{bmatrix}$$

$\Rightarrow n\sqrt{a} = n\sqrt{b} = 72 \Rightarrow a = b$ and $72a + 36\left(72 - \sqrt{a}\right) = 3600$

$\Rightarrow \sqrt{a} = 4 \Rightarrow a = 16$ and $n = 18$.

85. 20

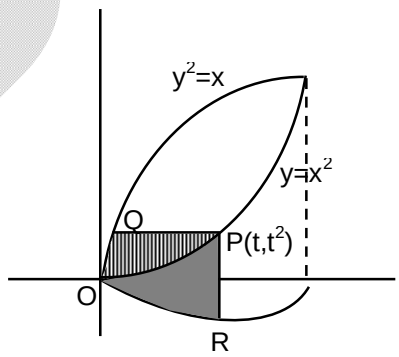
Sol. Let $P(t, t^2)$, $Q(t^4, t^2)$

$$\int_0^t \left(x^2 - f(x)\right)dx = 2\int_0^{t^2} \left(\sqrt{y} - y^2\right)dy$$

$\Rightarrow t^2 - f(t) = 2(t - t^4)2t$

$\Rightarrow f(t) = 4t^5 - 3t^2$

$$f\left(\frac{1}{2}\right) = \frac{4}{32} - \frac{3}{4} = -\frac{20}{32}.$$



86. 20

Sol. $f(x) = 1 + 4^x$

$$\lim_{x\to 1} \frac{\sin(4^x - 4) + 2 - 2^x}{x^3 - 7x^2 + ax + b} = \ln 8$$

For $\left(\frac{0}{0}\right)$ form $a + b = 6$

Also $\lim_{x\to 1} \frac{\left(\cos(4^x - 4)\right)4^x \ln 4 - 2^x \ln 2}{3x^2 - 14x + a} = \ln 8$

$$\Rightarrow \frac{6\ln 2}{a - 11} = 3\ln 2$$

$\Rightarrow a - 11 = 2 \Rightarrow a = 13, b = -7$

87. 81

Sol. For continuity at x = 0

$$\lim_{x\to 0}\frac{2^{\frac{cx}{3}}-1}{\frac{cx}{3}}\cdot\frac{c}{3}=\frac{\ln 4}{3}\Rightarrow c=2$$

For differentiability at x = 0

$$\lim_{x\to 0^-}\frac{6a(\ln 2)^2}{1+\left(\frac{b+x}{3}\right)^2}\cdot\frac{1}{3}=\lim_{x\to 0}\frac{\frac{2^{\frac{cx}{3}}-1}{x}-\frac{\ln 4}{3}}{x}$$

$$\Rightarrow \frac{6a(\ln 2)^2}{3\left(1+\frac{b^2}{9}\right)}=\lim_{x\to 0}\frac{3(2^{\frac{cx}{3}}-1)-x\ln 4}{3x^2}$$

$$=2\frac{(\ln 2)^2}{9}=\lim_{x\to 0}\frac{3\left(2^{\frac{cx}{3}}\ln 2\times\frac{c}{3}\right)-\ln 4}{6x}=\frac{2(\ln 2)^2}{9}$$

$$=\frac{6a}{3}\frac{(\ln 2)^2\times 9}{b^2+9}=\frac{2(\ln 2)^2}{9}=\frac{b^2+9}{a}=81$$

88. 1386

Sol. $h(g(g(x))) = x$

$h'(g(g(x)).g'(g(x).g'(x) = 1$; $g(x) = f^{-1}(x)$

$h'(0) \Rightarrow g(g(x)) = 0 \Rightarrow g(x) = f(0) = 3 \Rightarrow x = f(3) = 309$

Also $g'(f(x)) = \frac{1}{f(x)}$

$g'(3) = \frac{1}{f'(0)} = \frac{1}{3}$

$g'(309) = \frac{1}{f'(3)} = \frac{1}{462}$

$\Rightarrow h'(0) = \frac{1}{g'(3)g'(309)}$

89. 10648

Sol. L_1 is parallel to vector $2\hat{i}-\hat{j}-\hat{k}$ and it passes through point (13, –7, 0).

Equation of plane P is x + 5y – 3z + 22 = 0

$$V=\frac{1}{6}\begin{vmatrix}-22 & 0 & 0\\ 0 & -\frac{22}{5} & 0\\ 0 & 0 & \frac{22}{3}\end{vmatrix}=\frac{(22)^3}{90}.$$

90. 225

Sol. Let $\hat{x}\cdot\hat{y}=a$, $\hat{y}\cdot\hat{z}=b$, $\hat{x}\cdot\hat{z}=c$

$\left|\hat{x}+\hat{y}-\hat{z}\right|=\left|\vec{p}\right| \Rightarrow a-b-c=2$ …(1)

Also $c\hat{y}-a\hat{z}=q$ …(2)

and $c\hat{y}-b\hat{x}=r$ …(3)

$\because\ \vec{q}\cdot\hat{y}=5 \ \ \therefore c-ab=5$ …(4)

and $c-a=1$ …(5)

Hence $a=1, b=-3, c=2$